Healthcare Procurement
Spend Analytics Pipeline

Course: SYST52461 – Big Data Storage and Analysis

Term Project

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1. Introduction

1.1 Scenario Overview

This project models the procurement operations of a regional hospital network, an environment characterized by high-volume purchasing and complex supplier relationships. Hospitals routinely acquire a wide range of goods, from surgical consumables and pharmaceuticals to administrative supplies and facility maintenance materials. As noted in the original document, *"procurement teams need to know which vendors they rely on most… and where their budget is going across departments and product categories."*

These operational realities make healthcare procurement an ideal context for demonstrating a scalable analytics pipeline. The data naturally spans multiple interconnected entities, enabling the design of a multi-table medallion architecture that mirrors real enterprise data engineering practices.

1.2 Tables and Their Relationships

The dataset consists of four core tables, bronze_purchase_orders, bronze_vendors, bronze_line_items, and bronze_spend_taxonomy. Each representing a distinct component of the procurement lifecycle.

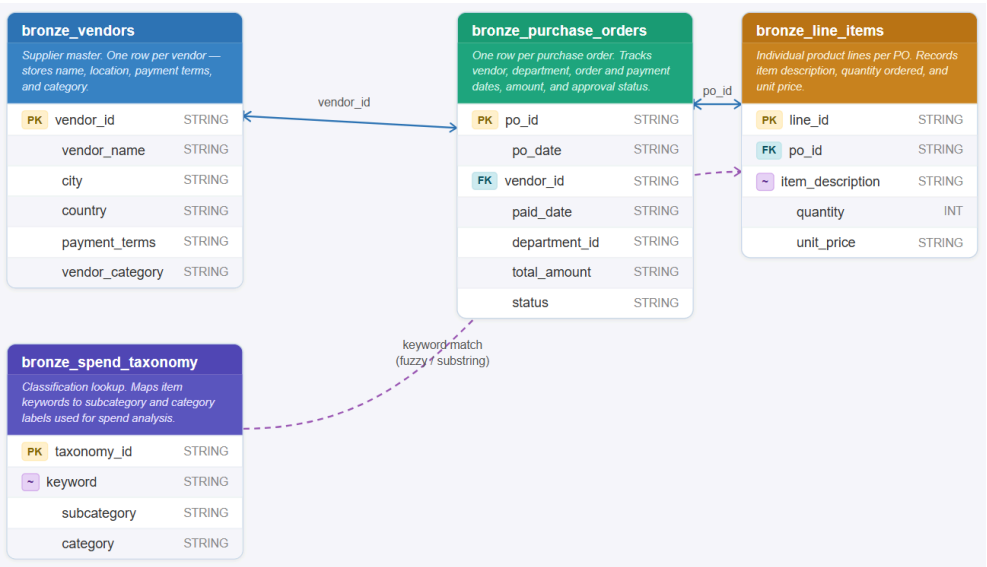

Figure 1: Bronze Layer - Entity Relationship Diagram

1.3 Why This Data Is Interesting

Procurement data offers several compelling analytical opportunities. Spend often clusters around a small group of suppliers, creating potential dependency risks. Payment timeliness is another important metric, as late payments can strain vendor relationships or lead to penalties. Finally, category-level spend analysis helps organizations understand where money is going and identify opportunities for consolidation or contract renegotiation.

These are the types of insights procurement teams typically monitor through dashboards, making this scenario a strong fit for the project.

2. Medallion Architecture

We implemented a three-layer medallion architecture in Databricks using PySpark and Delta tables. The Bronze layer captures raw data, the Silver layer cleans and standardizes it, and the Gold layer aggregates it into analysis-ready outputs.

2.1 Bronze Layer: Raw Ingestion

The Bronze layer stores the mock data exactly as generated, including all intentional imperfections. These issues mimic real-world inconsistencies, such as mixed date formats, inconsistent casing, missing vendor information, and irregular item descriptions.

We introduced the following intentional quality issues to simulate real-world data:

- **purchase_orders**: po_date and paid_date stored as strings in three mixed formats (YYYY-MM-DD, MM/DD/YYYY, and DD-MM-YYYY). total_amount stored as a string with dollar signs and commas (e.g. "$1,200.00"). Status labels inconsistently cased ("approved", "APPROVED", "aprvd"). About 5% of vendor_id values are null, and about 15% of paid_date values are null for Pending and Rejected orders.
- **vendors**: vendor_name contains mixed casing and spelling inconsistencies (e.g. "johnson & johnson" vs "JOHNSON AND JOHNSON"). payment_terms stored in multiple formats ("Net 30", "NET30", "n/30"). About 10% of city and country values are null.
- **line_items**: unit_price stored as a string with occasional dollar sign prefixes. About 3-5% of quantity values are zero or negative. About 2% of item_description values are null.
- **spend_taxonomy**: Some keywords duplicated with different casing. About 5% of subcategory and category values are null.

Keeping the Bronze layer raw and unmodified means that if a cleaning decision turns out to be wrong, we can always go back and reprocess from the original data.

2.2 Silver Layer: Cleaning and Standardization

In the Silver layer, I cleaned and standardized the raw Bronze tables so they could be used reliably in the later stages of the pipeline. The main goal here was to fix formatting issues, handle inconsistent values, remove invalid records, and make the data more structured without changing the business meaning of the original data.

Table-Specific Cleaning

- **Vendors:** Replaced missing city and country values with "Unknown" to retain records for spend analysis. Names were standardized for consistency, and payment terms were normalized into three standard values: Net 30, Net 45, and Net 60.

- **Purchase Orders:** Resolved mixed date formats using a coalesce parsing function. The total_amount was converted to **DoubleType** after removing currency symbols, and rows with null vendor_id were removed to ensure join integrity.
- **Line Items:** Filtered out invalid records with non-positive quantities or null descriptions. Text fields were trimmed and lowercased to improve the quality of future taxonomy matching.
- **Spend Taxonomy:** Deduplicated keywords using window functions and standardized category casing. A specific correction step was added to restore acronyms like **PPE** that were affected by automatic capitalization.

After cleaning each table, I validated the outputs by checking row counts, null values, distinct values, and remaining abnormal records before saving them as Delta tables. This step helped confirm that the Silver layer was clean, consistent, and ready for the Gold transformations.

2.3 Gold Layer: Aggregation and Analysis

The Gold layer joins and aggregates the Silver tables into analysis-ready tables. Each table answers a specific business question directly. This avoids further transformation in the dashboard or analysis notebook.

Processing workflow shows as below:

- **gold_spend_by_vendor**: This table identifies the most valuable medical suppliers. It joins approved purchase orders to vendors and calculates exact financial totals, order counts, and average order values per vendor. A spend_rank column is included to easily find top suppliers.
- **gold_spend_by_department**: This section tracks internal hospital budgets. It aggregates the total approved spend and order counts by department, showing exactly how much money each medical unit consumes.
- **gold_monthly_trend**: This table reveals purchasing patterns over time. It groups approved spending by year and month to help administrators see financial growth and seasonal changes clearly.
- **gold_payment_cycle**: This step analyzes hospital payment efficiency. It checks if the hospital pays its vendor bills on time by calculating the days taken to pay. It extracts the numerical limit from the payment terms (changing "Net 30" to 30) and flags any order that exceeds this agreed timeline.
- **gold_spend_by_taxonomy**: This table categorizes overall hospital spending to show what types of medical supplies cost the most money. It uses a keyword substring match to link raw item descriptions to official subcategories and categories. Items without a match are labeled "Unclassified".

A critical design decision in the taxonomy join is handling multiple keyword matches. A description like "Surgical Gloves (Box of 100)" matches both "gloves" and "surgical gloves". The code resolves this by keeping the longest matching keyword. The assumption is that a more specific match is more accurate.

3. EDA Findings

The following section presents three key insights from our exploratory data analysis. These findings are drawn from the Gold tables and are visualized in the Databricks dashboard.

3.1 Insight 1: Vendor Spend Is Highly Concentrated

A few vendors drive most of the gold spend, with one supplier dominating enough to signal a possible single-source dependency. This concentration is typical in healthcare, where long-term contracts and bulk purchasing often funnel spend to a small set of suppliers.

3.2 Insight 2: A Significant Share of Invoices Are Paid Late

Comparing gold_payment_cycle with days_to_pay, it shows many invoices were paid late, especially for Net 30 vendors. These delays risk penalties and strain relationships, suggest the accounts payable process may need stronger controls or automation.

3.3 Insight 3: Medical Supplies Dominate Category Spend

The taxonomy analysis shows Medical Supplies as the largest spend category, driven mainly by PPE and other consumables. Medical Equipment follows, with Pharmacy and Facilities further behind. PPE items like gloves and masks form the biggest subcategory, consistent with typical hospital purchasing patterns.

4. Reflection

4.1 Challenges We Faced

4.1.1 Taxonomy join complexity

The most technically challenging part of the project was the taxonomy join. Unlike the other joins which used exact foreign key matches, the taxonomy join required matching a free-text item description against a list of keywords using substring containment. When we ran this as a cross join (every line item against every keyword), it was expensive even on our relatively small dataset. We also had to handle the case where multiple keywords matched the same description and pick the most specific one using a window function. Getting this logic right took several iterations.

4.1.2 Mocking realistic data quality issues

Generating data that was messy in realistic ways required more thought than expected. It was easy to introduce random nulls and wrong types, but making the quality issues feel plausible, for example, having paid_date null only for Pending and Rejected POs rather than randomly, required building logic into the data generation step. We also had to ensure that our intentional quality issues were diverse enough to demonstrate a range of Silver cleaning techniques.

4.1.3 Date parsing with mixed formats

Handling the po_date and paid_date columns was a significant challenge due to the presence of multiple date formats within the Bronze layer.

- **Multi-Pattern Solution:** I developed a reusable function that applied several date patterns via a coalesce strategy, returning the first successful match to ensure accurate standardization.
- **Logic Preservation:** null values in paid_date were intentionally preserved, as they correctly represented **Pending** or **Rejected** orders rather than data errors.
- **Future Scalability:** While effective for this dataset, I noted that a production-grade pipeline would require a more flexible parsing strategy to handle a broader range of potential formats efficiently.

4.1.4 Coordinating dependent tasks across members

Since the four notebooks must run in sequence, each member had to wait for the previous notebook to finish and verify that the output was correct, checking row counts, null reports, and sample data, before starting their own work. Any error in one notebook would block everyone else. With more time, we would automate this using Databricks Jobs so the sequencing and validation happens without manual coordination.

4.2 Future Enhancements

4.2.1 Taxonomy Classification Upgrade

Future improvements include upgrading the taxonomy classification. The current keyword substring match misses typographical errors and abbreviations. A practical solution involves fuzzy matching with similarity scoring, and a machine learning classifier represents the ideal long term solution for a production system.

4.2.2 Pipeline Optimization

The data processing system should transition from a full refresh to incremental loading using the merge capability of Delta Lake. This approach ensures only new and changed records undergo processing during each run, rather than the entire dataset.

4.2.3 Data Quality

Replacing synthetic data with real procurement data will significantly improve the analysis. Synthetic data cannot completely replicate the unpredictable patterns and complex edge cases found in actual hospital purchasing systems.

4.2.4 Advanced Reporting

A vendor scorecard table is a valuable addition to the Gold layer. This specific table will combine spend concentration, payment compliance, and category coverage for each vendor, providing procurement managers with a single consolidated view of every supplier.

4.2.5 Automation

Finally, the entire pipeline requires automation using Databricks scheduled jobs. This configuration allows the final dashboard to refresh automatically without any manual notebook execution.